

Equity Risk Premium

Module 1 · Risk-free rate, ERP, country risk & implied ERP

Risk-free & FX

DCF basics

Cost of equity

Country risk

Lambda

Implied ERP

1. Golden rule for M&A valuation

IMPORTANT

When valuing a target company in M&A, always use the WACC or Ke of the target company, except in special cases like LBO, option pricing etc.

2. Risk-free rate & exchange rate

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EXAMPLE

An apple costs \$5 in the USA and ₹500 in India. Exchange rate: \$1 = ₹100. Inflation: 2% USA, 3% India.

	Today	After inflation
USA	\$5	$\$5 + 2\% = \5.10
India	₹500	$₹500 + 3\% = ₹515 (= \$5.15 \text{ if FX unchanged})$

USA is now cheaper → people buy from USA → demand for \$ rises → \$ appreciates vs ₹. To fix this, the rupee depreciates until prices match: $\$1 = 515 / 5.10 = ₹100.98$. This proves that the exchange rate follows the inflation differential.

NOTE

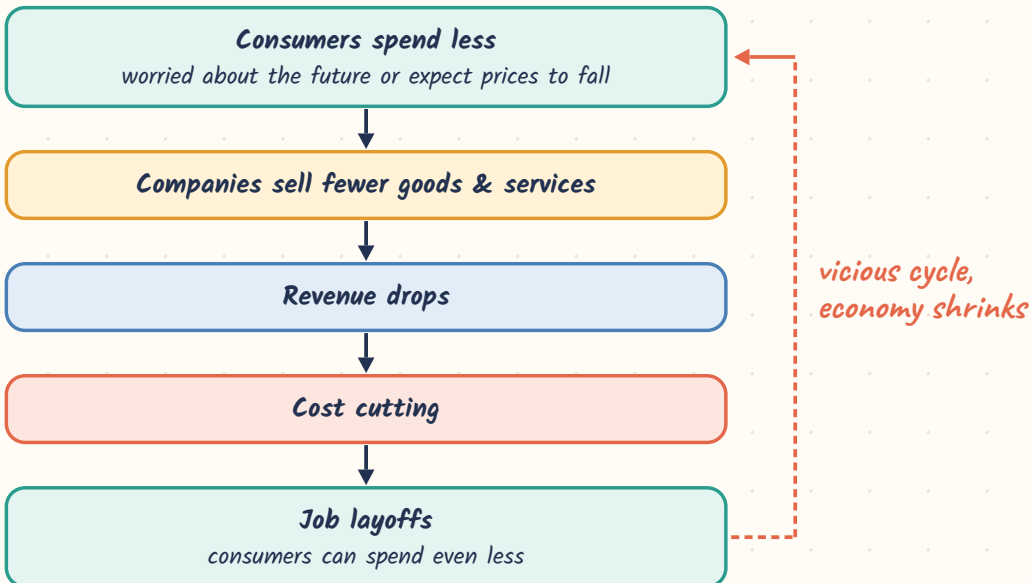
When valuing a company, it does not matter which currency you use, as long as the discount rate (risk-free rate) is in the same currency as the cash flows. The rate offsets the exchange rate.

e.g. Brazil 7.5% vs USA 2.5%: it looks like Brazil gives lower value if 7.5% is used, but the cash flows also carry that inflation and risk, so it evens out.

Nominal vs real rates

- We mostly use **nominal** rates, because inflation cannot be ignored. Ignoring inflation means cash flows must have constant pricing (prices never change).
- **Real** risk-free rates can be used where inflation is **volatile or uncertain**, but the world mostly operates in nominal terms.
- The **TIPS rate** gives the real risk-free rate and is assumed constant throughout the world.
- High inflation in the discount rate → **high growth** in cash flows. Low or negative inflation → **g falls much further**, reflecting the compensation.
- Bond / Treasury / Repo rates are driven by **inflation and real growth**.

Deflationary / shrinking economy (negative interest rates)

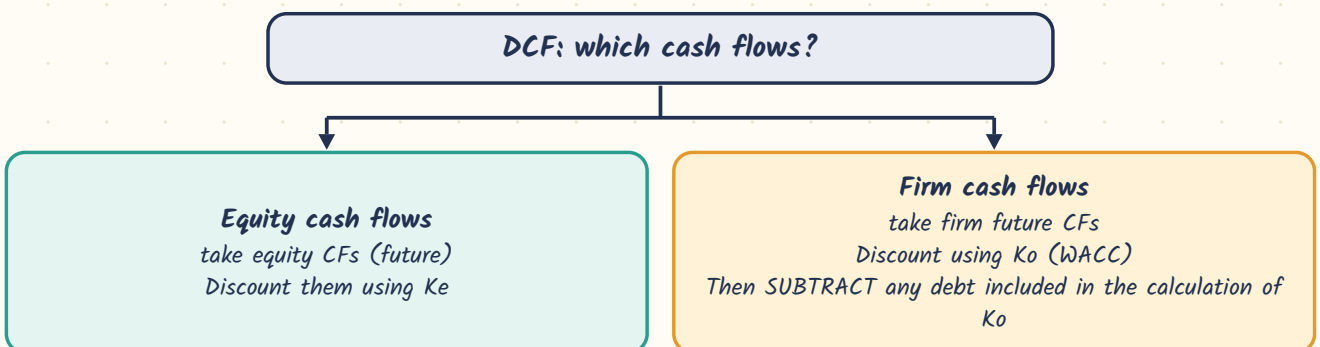


Building the risk-free rate for a currency

$$\text{Risk-free rate (currency)} = \text{US Treasury rate} + \text{Expected differential inflation}$$

- The differential method **strengthens** the assumed risk-free rate of a country that issues a govt bond, and acts as an **alternative** if the country does not issue one.
- Get the next 5 years of expected inflation from the **IMF website**.
- **Do not normalize risk-free rates.** If r_f is 1% and you average it to 5%, you are saying the stock is expensive, so you will not buy it, and you look for a better investment. But there is no 5% return in actual.

3. Two ways to do a DCF



4. Growth & terminal value

- Terminal value assumes the firm's CFs **grow at a steady rate forever**. It is logical to assume this constant rate is **not more than the nominal interest rate** in the terminal year.
- Top-line items generally grow slower than bottom-line items because of the **leverage effect**.
- Use **revenues** to estimate growth of a firm, because below-the-line items can be manipulated by cost cutting or other factors (look at **gross margin**).

- Check for **economies of scale**: is revenue growing faster than costs? (look at **operating margins**).
- The higher the **g**, the more you have to **reinvest**.

5. Estimating K_e : whose expectations count?

- Every investor has different expectations, so we do not look at all of them. Focus on the **marginal investor**, the one who actively buys and sells stocks and **affects market prices** (hedge funds, institutional buyers, mutual funds, pension funds).
- The risk of a company is **not its standalone risk** but the **risk it adds to a diversified portfolio**.
- Always use **market weights** in a WACC calculation.

NOTE

The marginal investor demands a return for **economic, macro and continuous risk**, because it is unavoidable and cannot be diversified away. Company-specific risk is ignored, since holding a large number of companies removes it.

In short: demands compensation for risk that affects all stocks.

TWO KINDS OF UNCERTAINTY

Estimation uncertainty can be reduced with more research and data. But **80-90% of uncertainty is economic uncertainty**, which cannot be removed.

6. Historical vs forward-looking premium

- The **historical premium** includes years of bad returns that distort future expectations.
- Instead use a **forward-looking premium**: what would you pay given the cash flows? Discount them using YTM / IRR to get the required return.

$$\text{Equity premium} = \text{IRR} - \text{Risk-free rate}$$

Problem with historical ERP: standard error

$$\text{Range} = \text{Average} \pm 2 \times \text{SE (95\% confidence)}$$

EXAMPLE

ERP (see table on page 48): $7\% \pm 2 \times 2.12 = 2.76\% \text{ to } 11.24\%$

That is an absurdly wide range, and we do not get a discrete value. Any average carries estimation error, so estimates should be given as a range.

△ Your handwritten notes say 4.76%. $7 - 4.24 = 2.76\%$, so I corrected it.

Also, we **cannot apply the US ERP to every country**, because their markets are not as mature as the US.

7. Country risk premium: 3 approaches

1 Default spread approach

Country risk is measured using the **government's default risk** (find it with the 3 methods discussed earlier).

$$\text{Country ERP} = \text{US ERP} + \text{Default spread}$$

Intuition: if the country's bond investors ask for an extra 2.83%, equity holders should demand **at least that much** extra.

Limitation: bond risk < equity risk, so this **underestimates equity risk**.

2 Equity volatility approach

Look at how volatile the country's stock market is compared to the US.

$$\begin{aligned}\text{Relative volatility} &= \sigma(\text{country}) / \sigma(\text{US}) \\ \text{ERP}(\text{country}) &= \text{ERP}(\text{US}) \times \text{Relative volatility}\end{aligned}$$

σ = standard deviation of market returns over roughly 2 years (100-104 weeks). This is a preferred approach, not a fixed number.

Intuition: reflects actual equity market behaviour. **Limitation:** ignores sovereign default risk.

3 Melded (preferred) approach

Combine both approaches, because **stocks are riskier than bonds**.

$$\text{Country ERP} = [\text{Default spread} \times (\sigma \text{ equity}(\text{country}) / \sigma \text{ bond}(\text{country}))] + \text{US ERP}$$

INTUITION

Bond investors only worry about **repayment**. Equity holders face micro and macro risk, profit risk, customer risk, competition and much more, so they should get a **bigger premium**. This approach includes the country's default risk and equity market risk.

IMPORTANT

How to get US ERP (implied approach): take the observed S&P 500 index point. Forecast future cash flows for the 500 companies in total. Use a reasonable terminal growth rate after some years. Find the **IRR** = return. Subtract the **10-year Treasury rate** → **US ERP**.

8. Operation-based CRP

IMPORTANT

Do **not** assign CRP by where a company is incorporated or headquartered. It should depend on **where the company operates and generates its revenues**.

Single emerging market: Embraer (2004)

Airplane manufacturer headquartered in Brazil. **97% of revenues** from USA & Europe, only **3% from Brazil**.

Assume $\text{ERP}(\text{mature}) = 5.00\%$, $\text{ERP}(\text{Brazil}) = 13.00\%$, $\text{CRP} = 8.00\%$.

Measure	Working	Result
Weighted ERP	$(0.97 \times 5.00\%) + (0.03 \times 13.00\%)$	5.24%
Weighted CRP	$(0.97 \times 0.00\%) + (0.03 \times 8.00\%)$	0.24%

It looks like a high-risk company, and inflating K_e due to its Brazilian incorporation would be **artificial**. Its revenue-weighted risk is barely higher than a US firm.

Multiple emerging markets: Ambev (2011)

Brazilian beverage company operating in Latin America and Canada. Rather than assigning 100% of the Brazil CRP, take the **weighted average CRP of all countries** where it sells its products (same process as above).

Multinational, regional breakdown: Coca-Cola (2012)

Coca-Cola rarely discloses revenue country-wise, only **region-wise**. Multiply each region's revenue share by that region's estimated average ERP and CRP. Result (2012 data): **total ERP = 7.14%, weighted CRP = 1.14%** (see slide 58).

PITFALLS OF REGIONAL DATA

1. Lumping safe and risky markets together.
2. Vague reporting categories like "Eurasia" and "Oceania".

Problems with revenue-based approaches

IMPORTANT

1. **Focus only on revenues.** It ignores where production, supply chain and physical assets are located. An emerging market firm might sell 90% in the US or Europe, but if 100% of its production, workforce and infrastructure sit in a politically unstable country, a revolution, nationalization or natural disaster could shut down production even if demand stays the same.
2. **Beta measures market volatility, not political risk.** Beta is market cyclicity (correlation between stock and market). Example: a company runs a gold mine in a politically unstable country. The govt could nationalize the mine, which raises risk, but $\beta < 1$ because gold is a stable-cost market.

NOTE

For a mature market, **ERP(mature)** does not need to be multiplied by λ . We assume no political / sovereign risk in the US (it can print dollars to pay lenders, so it never defaults). The only risk is market risk, so we multiply β with **ERP(mature)**. λ measures how vulnerable a specific company's cash flows are to political and sovereign risk.

9. Production-based ERP & lambda (λ)

Solution 1: Production-based ERP (Royal Dutch Shell, 2015)

For commodity, mining, oil & gas companies, customer location does not dictate risk (almost everyone needs the product) but the **source of physical extraction** does.

$$\text{Weighted ERP (production)} = \sum [\text{country share of total production} \times \text{that country's ERP}]$$

Solution 2: Estimate lambda (λ) for country risk

$$\text{Cost of equity} = R_f + (\beta \times \text{ERP(mature)}) + (\lambda \times \text{CRP})$$

Factors influencing λ :

- **Risk management & hedging** reduces risk exposure ($\lambda \downarrow$).
- **Industries vital to the govt's national interest** ($\lambda \uparrow$).

Revenue-based lambda

$$\lambda = \% \text{ of domestic revenue (firm)} / \% \text{ of domestic revenue (average firm)}$$

- Being incorporated in a country does **not** automatically mean the firm bears its CRP entirely.
- **Corporate strategy** (diversifying revenues internationally) can actively lower λ .

Price-return based lambda (regression approach)

For publicly traded companies this is a market-driven, statistical approach. It measures how sensitive a company's stock price is to sovereign crisis signals. Regress the company's stock returns against the returns of the country's sovereign bond (e.g. Brazilian C-Bond, denominated in USD).

$$\text{Return(company)} = \alpha + \lambda \times \text{Return(sovereign bond)} \quad [\lambda = \text{slope of the line}]$$

Company (2000–2003)	Regression	λ
Embraer	$\text{Return} = 0.0195 + 0.2681 \times \text{Return(C-Bond)}$	Low
Embratel	$\text{Return} = -0.0308 + 2.0030 \times \text{Return(C-Bond)}$	High

(See slide 63)

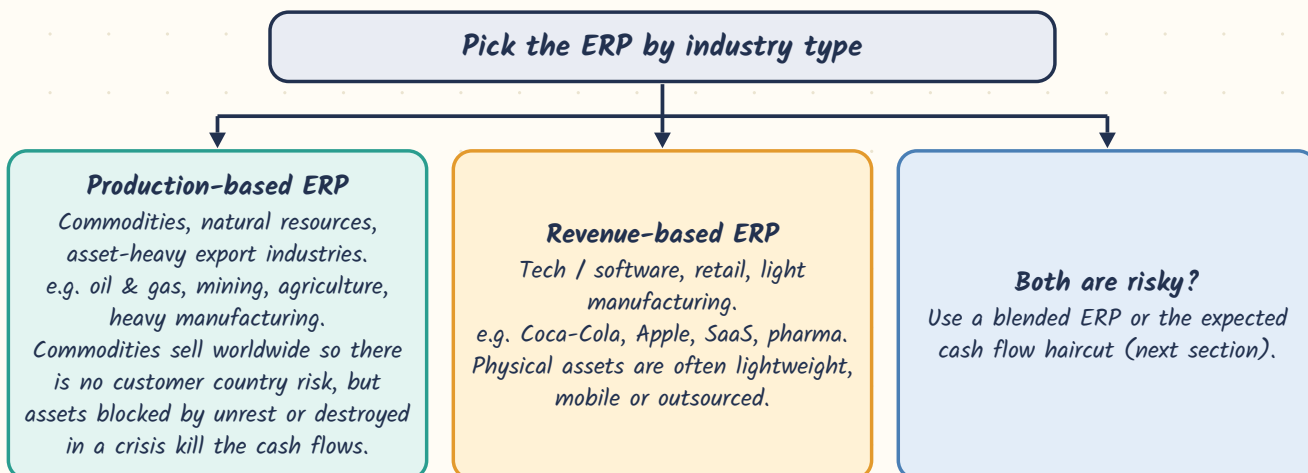
IMPORTANT

We do not use a weighted λ , since it is hard to execute (regression on massive data can include errors in assumptions). We take $\lambda(\text{home})$ calculated through the two methods above.

Priority: whether or not a country issues USD bonds, prioritise the **revenue-based** λ first and regression second, since the latter may look precise but can contain errors.

"It's better to be vaguely right than precisely wrong."

Which weighted ERP should I prioritise?



10. What if both revenue and production are risky?

Method 1: Dual-weighted blended ERP

$$\text{ERP}(\text{blended}) = \frac{(\text{in revenue} \times \text{ERP}(\text{revenue})) + (\text{in production} \times \text{ERP}(\text{production}))}{\text{in revenue} + \text{in production} = 1 \text{ [in = investment]}}$$

$$\begin{aligned} \text{in production} &= \text{Net PPE} / \text{Total firm assets (total capital invested)} \\ \text{in revenue} &= 1 - \text{in production} \end{aligned}$$

EXAMPLE

A company has \$80 million in factories (machines) and \$200 million total assets. $\text{in production} = 80/200 = 40\%$, $\text{in revenue} = 60\%$.

△ Your notes show 30% / 70%, but 80/200 gives 40% / 60%. Check the numbers from your lecture.

Method 2: Expected cash flow "haircut" (Damodaran's preferred)

Discount rates should only measure **continuous market risk**, while production catastrophes are **discrete event risks**. So instead of inflating the discount rate for production disruption, keep the discount rate anchored to the revenue ERP and apply **explicit probabilities directly to expected cash flows**.

$$\text{Expected CF} = (\text{Normal CF} \times P(\text{success})) + (\text{Distressed CF} \times P(\text{disruption}))$$

Three objective market and macroeconomic indicators can guess these probabilities reliably (vaguely):



1 Market-implied probability via sovereign CDS spread

If a country has an active CDS market, the financial market is already pricing the exact probability of a sovereign crisis / default.

$$P(\text{distress / disruption}) = \text{Sovereign CDS spread} / (1 - \text{Recovery rate})$$

Standard corporate finance assumes a **40% recovery rate** (0.40): investors expect to salvage 40 cents on the dollar in a sovereign default.

2 Credit rating agency cumulative default tables

If no CDS exists, look up the sovereign rating (Moody's, S&P or Fitch) and use their **historical cumulative default rate (CDR)** tables.

- Look up the country's sovereign rating.
- Pick the time horizon matching your CF forecast.
- The **cell value** is $P(\text{disruption})$.
- The sovereign debt should be **USD denominated**.

3 Quantitative political risk scores

Organizations like the **ICRG**, the **Economist Intelligence Unit (EIU)** and the **World Bank** (Governance Indicators) publish composite country risk indices scored 0 to 100.

$$\text{Risk score} = 100 - \text{Country safety index}$$

Risk category	Country safety index	Risk score	Implied P(disruption)
Very low risk	80 - 100	0 - 20	0% - 2%
Moderate risk	60 - 79	21 - 40	3% - 8%
High risk	40 - 59	41 - 60	9% - 18%
Extreme risk	0 - 39	61 - 100	> 20%

11. Implied equity risk premium

$$\text{Implied ERP} = \text{Expected return on equity} - \text{Risk-free rate}$$

Unlike historical ERP, it is **forward-looking**.

INTUITION

Every asset has a **current price** and **expected future CFs**. The **IRR** that equates both is the expected return on equity. Subtract the risk-free rate to get the ERP.

For stocks: **cash flows** = dividends + share buybacks; **price** = current market index.

Steps

Estimate cash flows (dividend + buyback) for the current year



Forecast CFs with an appropriate growth rate using analysts' estimates



Assume terminal growth ~ nominal GDP growth or the (terminal year) risk-free rate



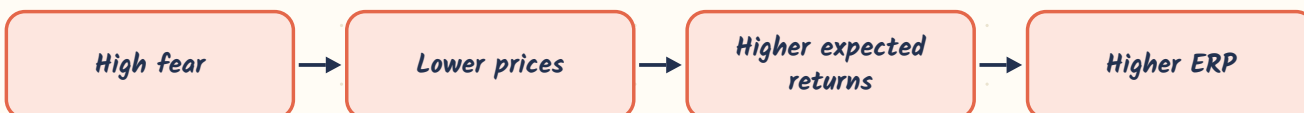
Calculate the IRR to get the expected return



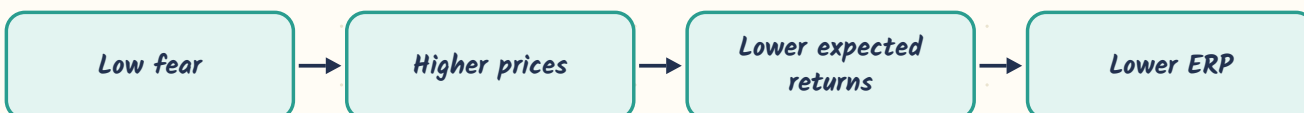
Implied ERP = Expected return - Risk-free rate

Implied ERP depends on: (a) current price, (b) expected CFs, (c) risk-free rate.

ERP is a market fear indicator



COVID crash: high fear → stock prices crashed → expected return ↑ → ERP ↑ (7.75%). To maintain the same IRR when the current price falls, we should have a higher IRR.



Dot-com bubble: low fear → stable, forever-growing stocks → lower expected returns → very low ERP (about 2%).

We assume expected CFs are not changing, because we look at one thing at a time (*ceteris paribus*).

$$\text{Expected return} = \text{Risk-free rate} + \text{ERP}$$

	Expected return	Risk-free rate	ERP
Jan 2020	7.12%	1.92%	5.20%
Jan 2025	8.91%	4.58%	4.33%

LEARNING (2025 vs 2020)

- Risk-free rate rose significantly as inflation rose after COVID, giving higher Treasury yields.
- ERP decreased slightly.
- Expected return stayed high, because it consists of both implied ERP and the risk-free rate.

- Implied ERP has fluctuated between **2% and 6.5%** over 1960-2024.
- Corporate default spread** (bond) measures credit risk = corporate bond yield – risk-free yield. **ERP** measures equity risk. During crises **both rise**, because the company's bonds and stock become riskier.

ERP vs real estate cap rates

$$\text{Cap rate} = \text{Net operating income (annual rent)} / \text{Property value}$$

If annual rent stays the same and property value increases, the cap rate falls (and vice versa). This is in line with ERP and CF: if rent is the same but one property is cheaper, it is probably in a riskier area (rent not paid on time, crime etc.). **High fear → lower prices, regardless of the rent being the same.**

NOTE

Current implied ERP has historically contained **more useful information about future returns** than the traditional historical ERP (see page 76: correlation of current implied ERP vs others).

Worked example: implied ERP for India, 5 Sep 2007 (page 77)

Sensex = 15,446. Dividend yield = 3.05% → $D_0 = 15,446 \times 3.05\% = 471.11$. Expected g for the next 5 years = 14% (high). Terminal growth = 6.76% (risk-free rate after 5 years).

Year	1	2	3	4	5
Cash flow	537.07	612.25	697.97	795.69	907.08

$$TV(5) = D6 / (r - g) = 907.08 \times 1.0676 / (r - 0.0676)$$

$$15,446 = \sum CF(t)/(1+r)^t + TV(5)/(1+r)^5$$

ANSWER

Solve for $r = 11.18\% \rightarrow$ Implied ERP = $11.18\% - 6.76\% = 4.42\%$

Emerging vs developed market K_e

$K_e(\text{emerging}) > K_e(\text{developed})$, because emerging markets carry additional risks: political risk, currency risk, economic instability, weaker institutions, lower liquidity, sovereign / credit risk.

$$\text{Differential} = K_e(\text{emerging}) - K_e(\text{developed})$$

For 2023: $8.93\% - 7.56\% = 1.37\%$. The risk gap between the two has generally narrowed over 2008-24 (page 78).

Which ERP to use, depending on your belief:

If you believe...	Use...
The market is reasonably priced	Current implied ERP
Today's ERP is unusual and will revert to norms (e.g. COVID 2020)	Historical ERP
The market is wrong in the short run but corrects over time (post-COVID)	Average implied ERP

IMPORTANT

If you start with the **developed market ERP**, then **add the differential** to get the emerging market ERP. **Do not do this if you have calculated the implied ERP** for the country, or you will double count country risk.